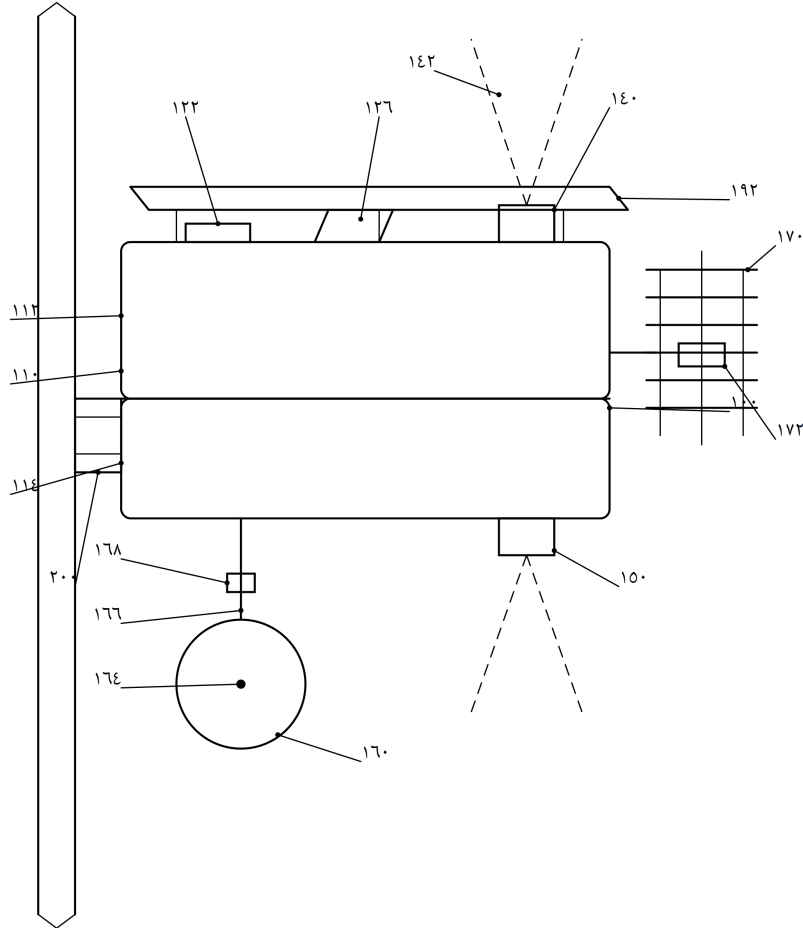


A heat sensor that knows when it is dirty

An instrument that measures how hot an outdoor space feels to a person, and corrects its own dust error using nothing but a second window — no lamp, no motor, no cleaning.



الشكل ١

What was filed

APPLICATION	FILED	OFFICE	STATUS
SA 1020268060	11 Sep 2026	SAIP • Saudi Arabia	Formal examination
CLAIMS	DRAWINGS	LANGUAGES	TERM IF GRANTED
20 • three independent	7 sheets	Arabic + English	20 years from filing

A compact instrument for the radiant thermal environment of outdoor space — built to be bolted to a pole, a bench, a lighting column or the underside of a shade canopy and then left alone, in dust, for years. It logs its own history, reports over a network, and measures and removes its own soiling error without a moving part, an internal emitter, or anyone ever cleaning it.

THE PROBLEM

Air temperature is the published number, and it is the wrong one

On a summer afternoon in Riyadh a weather station reports 43 °C. A person standing on that pavement is absorbing far more than the air is carrying: direct and reflected solar radiation, long-wave emission from hot ground and façades, and — where a shade structure stands overhead — radiation from its underside. Mean radiant temperature, not air temperature, is what decides whether an outdoor space is usable in the afternoon.

Saudi Arabia is building shade at scale. Once a canopy is up, there is no instrument in the field that can say how much of the remaining discomfort that canopy is responsible for, or whether a lighter, higher or ventilated one would do better.

The instruments that could answer this have a second problem. Every one of them reads the sky through an optical aperture, and in an arid climate that aperture silently loses transmittance to settled dust — measurably within weeks, with no rainfall between June and September to wash it off. The instrument does not report that it is dirty. It reports a falling number, indistinguishable from a genuinely dimmer sky.

Two optical windows, neither of them ever cleaned

Prior-art soiling compensation takes one of four forms: an internal emitter tested against a stored clean-state baseline; a motorised cover; a second reference instrument that is automatically cleaned; or an independent measurement that is subtracted out. Three of the four add either an active emitter or a moving part — the components least likely to survive in the environment they were added to survive.

This apparatus carries **two optical channels whose apertures soil at deliberately different rates** — by a difference in inclination, in bore depth-to-diameter ratio, or in surface treatment. Both are left to soil. Neither is ever cleaned, and no clean reference exists anywhere in the system.

The two channel signals are $S_1 = k_1 \cdot r_1 \cdot t_1 \cdot E$ and $S_2 = k_2 \cdot r_2 \cdot t_2 \cdot E$, where E is the incident irradiance, k and r are fixed instrumental constants, and t is the instantaneous transmittance of each aperture. Their ratio, $R = (k_1 r_1 t_1) / (k_2 r_2 t_2)$, has E cancelled out of it. That cancellation is the invention: the ratio is blind to cloud and to absolute calibration, and responds only to the two apertures soiling at different rates. Fitting the divergence of R from its clean value recovers a common soiling transmittance factor, which is then applied to correct the radiometric output.

THE MECHANISM IN FOUR LINES

Clean	100 and 100 — the two channels hold a fixed relationship.
Cloud	50 and 50 — both fall together. The ratio is unchanged, so no correction is applied.
Dust	80 and 90 — they diverge. Cloud cannot do this; only unequal soiling can.
Correction	The flat aperture soils at twice the rate of the inclined one, so the 10-unit gap implies 10 lost from one and 20 from the other. Reported irradiance: 100 , not 80.

A step recovery of the ratio toward its reference value identifies a rainfall event, and the cumulative deposition variable resets itself without intervention. Where the recovered deposition exceeds the range in which the fit is valid, the processor withholds the correction, flags the affected channels as unreliable and transmits a service request. The marginal cost of the second channel is one photodiode.

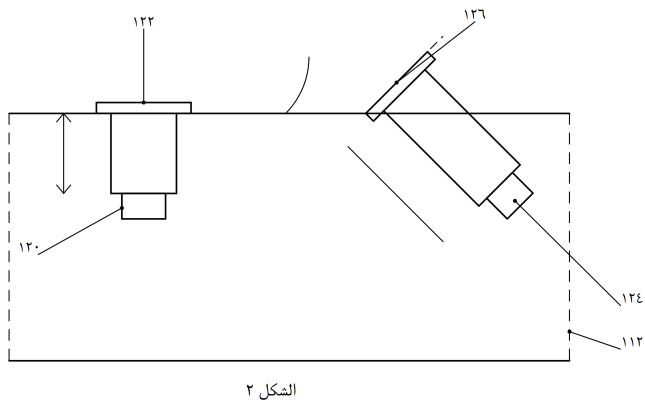


Fig. 2 — The first aperture (122) lies substantially horizontal; the second (126) is inclined, so dust settles on it more slowly.

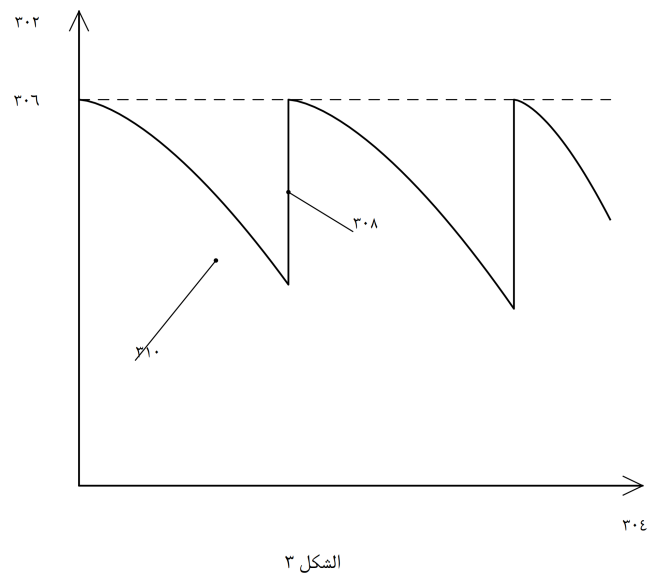


Fig. 3 — The ratio drifts as the apertures soil unequally. Each vertical recovery is a rainfall event detected and reset automatically.

Attributing radiant load to the structure that caused it

Existing instruments report one aggregate figure for mean radiant temperature. Where a shade structure stands overhead, its underside occupies a large fraction of the upper hemisphere a person exchanges radiation with, and in service it can run far above air temperature. The aggregate figure cannot separate that contribution from the ground, the façades or the sky — so the party who paid for the canopy cannot tell from it whether the canopy is working.

An upward infrared channel with a **bounded field of view** — a half-angle of not more than 35° , preferably 10° to 25° — is filled by the underside of the structure, and therefore reports that surface's temperature and substantially nothing else. A view factor is computed with respect to a reference human body at a defined height, and from it a separable attribution term: the share of mean radiant temperature for which that specific structure is responsible.

Dependent claims convert the result into the number of usable hours per day, and report how that number would change under a stated modification to the structure's emissivity, height or ventilation. The output stops being a complaint about a location and becomes a specification for a remedy. No image is formed at any stage.

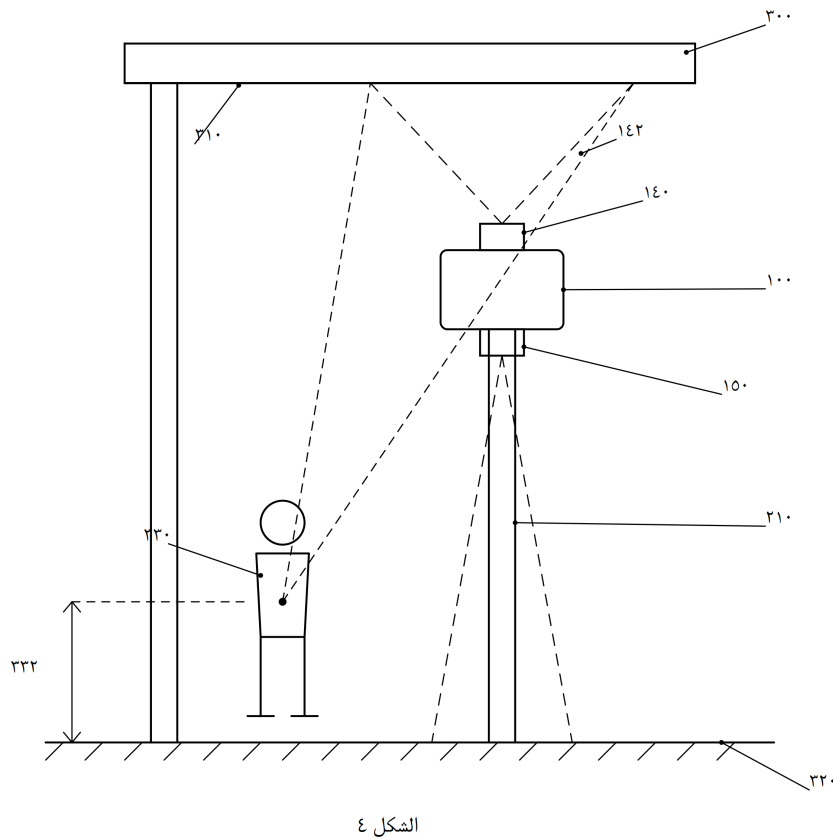


Fig. 4 — The bounded field of view (142) of the upward channel is filled by the underside (310) of the overhead structure (300). The view factor is taken with respect to a reference body (330), with the ground (320) contributing separately.

Measuring the thermometer's own error instead of assuming it away

Mean radiant temperature is classically derived from a globe thermometer — a sphere of known emissivity at equilibrium between radiant gain and convective loss. Full-size globes are 150 mm and slow; reduced-diameter globes respond faster and cost less, but the sensor must be carried inside on a support, and that support conducts heat between the sphere's interior and the ambient air. Because mean radiant temperature is derived from the *difference* between globe and air temperature, and that difference is small in a small globe, the resulting bias is not negligible. The published literature on globe error addresses convection, not conduction along the probe support.

Here a **second temperature sensor sits on the probe support itself**, spaced from the sphere wall. The gradient between it and the central sensor measures the conductive flow directly, and the correction follows from the known conductance of the support. A dependent claim estimates the sphere's time constant from the same pair of readings and applies a dynamic correction for lag under rapidly varying insolation.

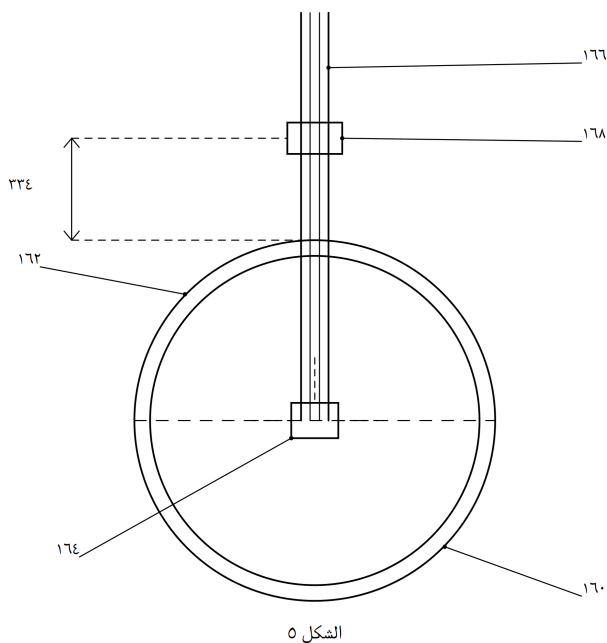


Fig. 5 — The sphere (162), the central sensor (164), the probe support (166) and the support-mounted sensor (168) from which conduction is measured.

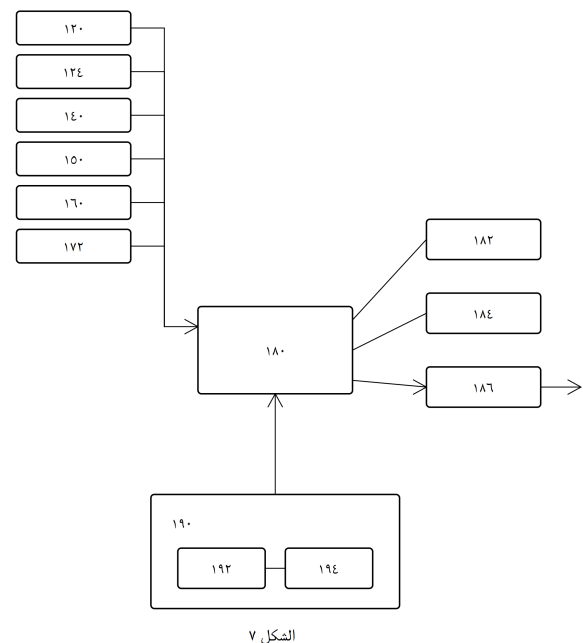


Fig. 7 — Architecture. The store (184) retains the corrected value, the uncorrected value *and* the factor applied, so history stays re-processable and auditable.

What already exists, and why the claims sit outside it

The claims were drafted after a search of the American, European and Gulf collections. Two documents come close enough to matter, and the specification cites both — disclosing known prior art is required, and a claim drafted around a document is far harder to invalidate than one drafted in ignorance of it.

DOCUMENT	WHAT IT CLAIMS	WHY THIS IS OUTSIDE IT
US 12,718,553 B2 Arizona Board of Regents granted 25 Aug 2026	Biometeorological sensing device — humidity, UV, anemometer, air and globe thermometers and a camera ; the processor estimates mean radiant temperature, bounds a detected person, generates a shadow map and computes their intersection.	Requires a camera. This apparatus has none, deliberately: imaging in public space is separately regulated, and occupancy here is sensed by a non-imaging millimetre-wave channel.
US 12,270,708 B2 Princeton University granted 8 Apr 2025	Mean radiant temperature sensor with occupancy detection, counting and tracking, carried on a scanning motion system of two or more degrees of freedom.	Requires a scanning mechanism. This apparatus has no moving parts at all.
US 11,774,282 B2 soiling detection	Pyranometer dome soiling detected by an internal LED, a diffusor and in-cavity detectors compared against a stored clean-state baseline.	Requires an emitter and a stored clean state. Claim 1 here expressly excludes both, and no clean state is ever retained.
Saudi and GCC collections	Searched for outdoor thermal comfort sensing.	No results.

Why it is a combination and not a collection

A patent is not granted for assembling known parts that each go on doing their own job. The test is whether the combination produces an effect no part produces alone — and here it does. Neither optical channel can, by itself, distinguish a dimmer sky from a dirtier window: a single channel sees one falling number and cannot say which. Only the ratio between two channels that soil at different rates carries that information. The effect lives in the relationship, not in either component.

One sensing core, seven mounting interfaces

The expensive part — the sensing core, its calibration and its certification — is built once. What changes between a bus shelter, a park bench, a lighting column and a canopy soffit is a bracket. Claim 10 covers the interchangeable interface; the specification describes seven adapters, including a module that seats inside a lighting-column enclosure.

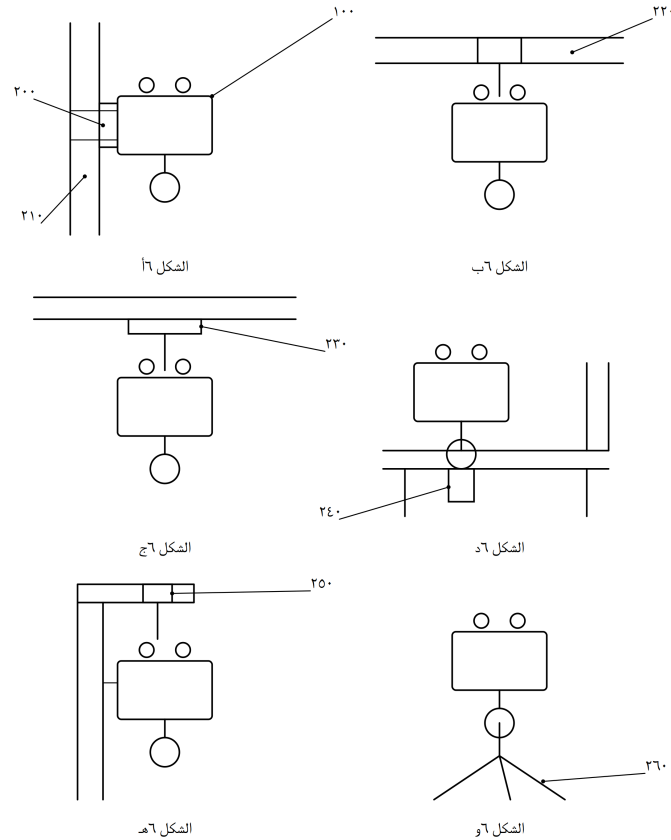


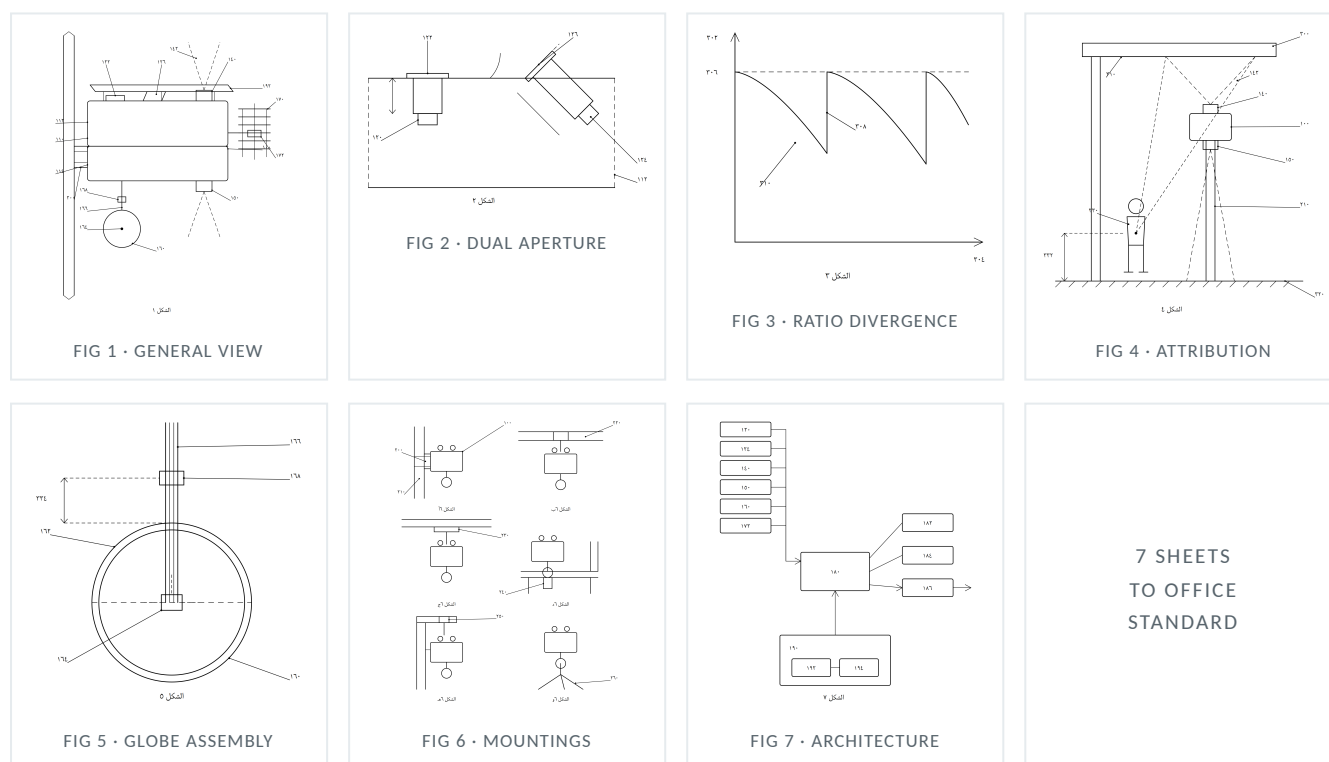
Fig. 6 — Mounting configurations 6A–6F: vertical pole, horizontal rail, canopy soffit, public seating, lighting column, freestanding base.

What the application contains

The specification was written to SAIP's content guide — its margins, its requirement that each content begin on a new page with its name centred and underlined, line numbering on every fifth line, the prohibition on illustrations inside the text documents, and the drawing standard covering sheet margins, reference-numeral height and caption placement.

CONTENT	ARABIC	ENGLISH
Abstract	132	167
Background and prior art	915	1,051
General description of the invention	510	681
Brief description of the drawings	122	181
Detailed description — ten sections	2,742	3,518
Claims — twenty	1,031	1,312
Words	5,452	6,910

Arabic is the mandatory language of filing; the English set was prepared alongside it. The detailed description closes with the best method known to the inventor at the date of filing, as the Regulations require, and enumerates nineteen further sensing channels the same core may carry — barometric, particulate, gas, acoustic, precipitation, soil and vibration among them — stating expressly that the list is not exhaustive.



Status and what happens next

11 Sep 2026	Filed at SAIP. Application No. SA 1020268060. Filing date secured — under Royal Decree M/27 Article 5(d), the patent goes to the earlier application, not the earlier inventor.
Sep 2026	First formality examination. Formal requirements under response.
Sep 2027	End of the Paris Convention priority year — the decision point for international filing (PCT or direct) claiming this priority date.
≈ Mar 2028	Publication at 18 months from filing.
2029–2031	Substantive examination on novelty, inventive step and industrial applicability. Current office figures put Saudi filing-to-grant at three to five years.

The application is pending. Grant is not assured, and the argument an examiner is most likely to test is not novelty but inventive step — whether substituting a differently-soiling aperture for a cleaned reference is obvious. The answer in the specification is that every prior method preserves, restores or stores a known clean state, and this one never has one: it recovers an absolute transmittance from a purely relative drift.

WHAT THE WORK INVOLVED

Scope

Instrumentation physics	Mean radiant temperature from globe thermometry; ISO 7726 and the Thorsson outdoor convection correction; view factors; wet-bulb globe temperature and UTCI as downstream indices.
Error budgeting	Identifying aperture soiling and probe-support conduction as the dominant field error terms, and designing a measurable correction for each rather than a tolerance for it.
Prior-art analysis	Reading granted claims in the US, European and Gulf collections and drafting around them, including the deliberate exclusion of a camera on regulatory grounds.
Claim architecture	Twenty claims across three independent lines, so that the failure of one does not take the others with it; negative limitations used to carve out the closest art.
Technical drawing	Seven sheets generated to the office drawing standard — margins, numeral height, unshaded line work, caption placement — with consistent reference numerals across all figures.
Regulatory process	Self-filed in Arabic through the SAIP e-services portal without an attorney, including the fee schedule, the content guide and the formality-response procedure.

Abdulah Alqabbani · abdulahalqabbani@gmail.com

Patent application SA 1020268060, Saudi Authority for Intellectual Property, filed 11 September 2026. Patent pending; no grant has been made. Figures reproduced from the drawing sheets as filed. This document is a summary — the claims as filed define the scope of protection.